

6 Linear Hyperbolic Systems

In this chapter we begin the study of systems of conservation laws by reviewing the theory of a constant coefficient linear hyperbolic system. Here we can solve the equations explicitly by transforming to characteristic variables. We will also obtain explicit solutions of the Riemann problem and introduce a “phase space” interpretation that will be very useful in our study of nonlinear systems.

Consider the linear system

$$\begin{aligned}u_t + Au_x &= 0 \\ u(x, 0) &= u_0(x)\end{aligned}\tag{6.1}$$

where $u : \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}^m$ and $A \in \mathbb{R}^{m \times m}$ is a constant matrix. This is a system of conservation laws with the flux function $f(u) = Au$. This system is called **hyperbolic** if A is diagonalizable with real eigenvalues, so that we can decompose

$$A = R\Lambda R^{-1}\tag{6.2}$$

where $\Lambda = \text{diag}(\lambda_1, \lambda_2, \dots, \lambda_m)$ is a diagonal matrix of eigenvalues and $R = [r_1 | r_2 | \dots | r_m]$ is the matrix of right eigenvectors. Note that $AR = R\Lambda$, i.e.,

$$Ar_p = \lambda_p r_p \quad \text{for } p = 1, 2, \dots, m.\tag{6.3}$$

The system is called **strictly hyperbolic** if the eigenvalues are distinct. We will always make this assumption as well. (For a demonstration that the behavior can change dramatically in the nonstrictly hyperbolic case, see Exercise 7.4.)

6.1 Characteristic variables

We can solve (6.1) by changing to the “characteristic variables”

$$v = R^{-1}u.\tag{6.4}$$

Multiplying (6.1) by R^{-1} and using (6.2) gives

$$R^{-1}u_t + \Lambda R^{-1}u_x = 0 \quad (6.5)$$

or, since R^{-1} is constant,

$$v_t + \Lambda v_x = 0. \quad (6.6)$$

Since Λ is diagonal, this decouples into m independent scalar equations

$$(v_p)_t + \lambda_p(v_p)_x = 0, \quad p = 1, 2, \dots, m. \quad (6.7)$$

Each of these is a constant coefficient linear advection equation, with solution

$$v_p(x, t) = v_p(x - \lambda_p t, 0). \quad (6.8)$$

Since $v = R^{-1}u$, the initial data for v_p is simply the p th component of the vector

$$v(x, 0) = R^{-1}u_0(x). \quad (6.9)$$

The solution to the original system is finally recovered via (6.4):

$$u(x, t) = Rv(x, t). \quad (6.10)$$

Note that the value $v_p(x, t)$ is the coefficient of r_p in an eigenvector expansion of the vector $u(x, t)$, i.e., (6.10) can be written out as

$$u(x, t) = \sum_{p=1}^m v_p(x, t)r_p. \quad (6.11)$$

Combining this with the solutions (6.8) of the decoupled scalar equations gives

$$u(x, t) = \sum_{p=1}^m v_p(x - \lambda_p t, 0)r_p. \quad (6.12)$$

Note that $u(x, t)$ depends only on the initial data at the m points $x - \lambda_p t$, so the domain of dependence is given by $\mathcal{D}(\bar{x}, \bar{t}) = \{x = \bar{x} - \lambda_p \bar{t}, p = 1, 2, \dots, m\}$.

The curves $x = x_0 + \lambda_p t$ satisfying $x'(t) = \lambda_p$ are the “characteristics of the p th family”, or simply “ p -characteristics”. These are straight lines in the case of a constant coefficient system. Note that for a strictly hyperbolic system, m distinct characteristic curves pass through each point in the x - t plane. The coefficient $v_p(x, t)$ of the eigenvector r_p in the eigenvector expansion (6.11) of $u(x, t)$ is constant along any p -characteristic.

6.2 Simple waves

We can view the solution as being the superposition of m waves, each of which is advected independently with no change in shape. The p th wave has shape $v_p(x, 0)r_p$ and propagates with speed λ_p . This solution has a particularly simple form if $v_p(x, 0)$ is constant in x for all but one value of p , say $v_p(x, 0) \equiv c_p$ for $p \neq i$. Then the solution has the form

$$\begin{aligned} u(x, t) &= \sum_{p \neq i} c_p r_p + v_i(x - \lambda_i t, 0) r_i \\ &= u_0(x - \lambda_i t) \end{aligned} \quad (6.13)$$

and the initial data simply propagates with speed λ_i . Since $m - 1$ of the characteristic variables are constant, the equation essentially reduces to $u_t + \lambda_i u_x = 0$ which governs the behavior of the i th family. Nonlinear equations have analogous solutions, called “simple waves”, in which variations occur only in one characteristic family (see [97]).

6.3 The wave equation

The canonical example of a hyperbolic PDE is the second order scalar wave equation,

$$u_{tt} = c^2 u_{xx}, \quad -\infty < x < \infty \quad (6.14)$$

with initial data

$$\begin{aligned} u(x, 0) &= u_0(x) \\ u_t(x, 0) &= u_1(x). \end{aligned}$$

This can be rewritten as a first order system of conservation laws by introducing

$$v = u_x \quad \text{and} \quad w = u_t$$

Then $v_t = w_x$ by equality of mixed partials, and (6.14) becomes $w_t = c^2 v_x$, so we obtain the system

$$\begin{bmatrix} v \\ w \end{bmatrix}_t + \begin{bmatrix} -w \\ -c^2 v \end{bmatrix}_x = 0, \quad (6.15)$$

which is of the form (6.1) with

$$A = \begin{bmatrix} 0 & -1 \\ -c^2 & 0 \end{bmatrix}. \quad (6.16)$$

The initial conditions become

$$\begin{aligned} v(x, 0) &= u'_0(x) \\ w(x, 0) &= u_1(x). \end{aligned} \quad (6.17)$$

Note that this system is hyperbolic since the eigenvalues of A are $\pm c$, which are the wave speeds.

EXERCISE 6.1. *Compute the eigenvectors of A in (6.16) and use these to decouple (6.15) into a pair of scalar equations. Solve these equations to find that*

$$v(x, t) = \frac{1}{2} \left[u'_0(x_1) + \frac{1}{c} u_1(x_1) + u'_0(x_2) - \frac{1}{c} u_1(x_2) \right] \quad (6.18)$$

$$w(x, t) = \frac{1}{2} [cu'_0(x_1) + u_1(x_1) - cu'_0(x_2) + u_1(x_2)] \quad (6.19)$$

where $x_1 = x + ct$, $x_2 = x - ct$. Use this to compute $u(x, t)$ in the original wave equation (6.14).

EXERCISE 6.2. *Rewrite the 2D wave equation*

$$u_{tt} = c^2(u_{xx} + u_{yy}) \quad (6.20)$$

as a first order system of the form $q_t + Aq_x + Bq_y = 0$ where $q = [v, w, \varphi]$ by introducing $v = u_x$, $w = u_y$, $\varphi = u_t$.

EXERCISE 6.3. *Solve the “linearized shallow water equations”,*

$$\begin{bmatrix} u \\ \varphi \end{bmatrix}_t + \begin{bmatrix} \bar{u} & 1 \\ \bar{\varphi} & \bar{u} \end{bmatrix} \begin{bmatrix} u \\ \varphi \end{bmatrix}_x = 0 \quad (6.21)$$

where $\bar{u}$ and $\bar{\varphi} > 0$ are constants, with some given initial conditions for u and φ .

Propagation of singularities. If we consider a point $(\bar{x}, \bar{t})$ for which the initial data is smooth at each point $x_0 = \bar{x} - \lambda_p \bar{t}$, $p = 1, 2, \dots, m$, then $v(x - \lambda_p t, 0) = R^{-1}u(x - \lambda_p t, 0)$ is also a smooth function at each of these points, and hence so is $u(x, t)$. It follows that any singularities in the initial data can propagate only along characteristics, just as in the scalar linear case. Moreover, smooth initial data gives smooth solutions.

6.4 Linearization of nonlinear systems

Now consider a nonlinear system of conservation laws

$$u_t + f(u)_x = 0, \quad (6.22)$$

where $u : \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}^m$ and $f : \mathbb{R}^m \rightarrow \mathbb{R}^m$. This can be written in the quasilinear form

$$u_t + A(u)u_x = 0 \quad (6.23)$$

where $A(u) = f'(u)$ is the $m \times m$ Jacobian matrix. Again the system is **hyperbolic** if $A(u)$ is diagonalizable with real eigenvalues for all values of u , at least in some range

where the solution is known to lie, and strictly hyperbolic if the eigenvalues are distinct for all u .

We can define characteristics as before by integrating the eigenvalues of $A(u)$. There are m characteristic curves through each point. The curves $x(t)$ in the p th family satisfy

$$\begin{aligned} x'(t) &= \lambda_p(u(x(t), t)), \\ x(0) &= x_0, \end{aligned}$$

for some x_0 . Note that the λ_p now depend on u , the solution of the problem, and so we can no longer solve the problem by first determining the characteristics and then solving a system of ODEs along the characteristics. Instead a more complicated coupled system is obtained, reducing the effectiveness of this approach. Characteristics do yield valuable information about what happens locally for smooth data, however. In particular, if we linearize the problem about a constant state $\bar{u}$ we obtain a constant coefficient linear system, with the Jacobian frozen at $A(\bar{u})$. This is relevant if we consider the propagation of small disturbances as we did for the scalar traffic flow model in Chapter 4. Assume an expansion of the solution with the form

$$u(x, t) = \bar{u} + \epsilon u^{(1)}(x, t) + \epsilon^2 u^{(2)}(x, t) + \cdots, \quad (6.24)$$

where $\bar{u}$ is constant and ϵ is small, then by the same derivation as equation (4.12) we find that $u^{(1)}(x, t)$ satisfies

$$u_t^{(1)}(x, t) + A(\bar{u})u_x^{(1)}(x, t) = 0. \quad (6.25)$$

Small disturbances thus propagate (approximately) along characteristic curves of the form $x_p(t) = x_0 + \lambda_p(\bar{u})t$. Higher order corrections for nonlinear problems can be obtained by retaining more terms in the expansion. The propagation of discontinuities and other singularities can also be studied through the use of similar expansions. This “geometrical optics” approach to studying weakly nonlinear phenomenon is a powerful tool, but one we will not pursue here. (See Whitham[97], for example.)

Weak shocks. Recall that for a linear system singularities propagate only along characteristics. For nonlinear problems this is not the case, as we have already seen for nonlinear scalar equations in Chapter 3. Instead, the Rankine-Hugoniot jump condition (3.33) must be satisfied for a propagating discontinuity,

$$f(u_l) - f(u_r) = s(u_l - u_r) \quad (6.26)$$

where s is the propagation speed. However, for very weak discontinuities this relates nicely to the linear theory. Suppose that

$$\|u_r - u_l\| \equiv \epsilon \ll 1. \quad (6.27)$$

In this case expanding $f(u_l)$ about u_r gives

$$f(u_l) = f(u_r) + f'(u_r)(u_l - u_r) + O(\epsilon^2) \quad (6.28)$$

so that (6.26) yields

$$f'(u_r)(u_l - u_r) = s(u_l - u_r) + O(\epsilon^2). \quad (6.29)$$

In the limit as $\epsilon \rightarrow 0$, the normalized vector $(u_l - u_r)/\epsilon$ must approach an eigenvector of $f'(u_r) = A(u_r)$ with s approaching the corresponding eigenvalue. This observation will be very important in studying the structure of nonlinear solutions more generally.

6.4.1 Sound waves

A familiar example of small disturbances in gas dynamics is the propagation of sound waves in the air. If we consider the Euler equations for a polytropic gas, in which the flux is (2.14) with equation of state (5.22), then the flux Jacobian matrix is

$$f'(u) = \begin{bmatrix} 0 & 1 & 0 \\ \frac{1}{2}(\gamma - 3)v^2 & (3 - \gamma)v & (\gamma - 1) \\ \frac{1}{2}(\gamma - 1)v^3 - v(E + p)/\rho & (E + p)/\rho - (\gamma - 1)v^2 & \gamma v \end{bmatrix}. \quad (6.30)$$

A tedious calculation confirms that the eigenvalues are given by $\lambda_1 = v - c$, $\lambda_2 = v$, and $\lambda_3 = v + c$, where

$$c = \sqrt{\gamma p / \rho}. \quad (6.31)$$

This is the local **speed of sound** in the gas. If we linearize these equations about some state $\bar{u}$, we see that small disturbances propagate at speeds $\bar{v}$, $\bar{v} \pm \bar{c}$, i.e. at speeds 0, $\pm \bar{c}$ relative to the background velocity $\bar{v}$. The waves traveling at speeds $\pm \bar{c}$ are sound waves; our ears are sensitive to the small variations in pressure in these waves.

The waves that travel with the velocity of the gas turn out to be simply density variations which advect with the fluid just as in the scalar linear advection equation. Even for the full nonlinear equations it is easy to check that there are solutions of the form

$$\begin{aligned} \rho(x, t) &= \hat{\rho}(x - \bar{v}t) \\ v(x, t) &= \bar{v} \\ p(x, t) &= \bar{p} \end{aligned} \quad (6.32)$$

where $\hat{\rho}(x)$ is any density distribution.

EXERCISE 6.4. *Verify that (6.32) satisfies the nonlinear Euler equations. Is this solution isentropic?*

EXERCISE 6.5. The sound speed for the Euler equations is most easily calculated using the equations in the form (5.26), where the entropy S is one of the variables and the equation of state gives $p = p(\rho, S)$. Linearize these equations about a constant state and compute the eigenvalues of the resulting matrix. This verifies that more generally the sound speed is given by $\sqrt{\partial p / \partial \rho}$, which for the polytropic case reduces to (6.31).

EXERCISE 6.6. Show that the sound speed in the isentropic equations (5.30) is the same as in the full Euler equations. For this system of two equations, small disturbance waves travel with speeds $\bar{v} \pm \bar{c}$. Waves of the form (6.32) traveling with speed $\bar{v}$ are not isentropic and no longer appear.

EXERCISE 6.7. Verify that linearizing the shallow water equations (5.38) gives the system (6.21). What is the “sound speed” for this system?

6.5 The Riemann Problem

For the constant coefficient linear system, the Riemann problem can be explicitly solved. We will see shortly that the solution to a nonlinear Riemann problem has a simple structure which is quite similar to the structure of this linear solution, and so it is worthwhile studying the linear case in some detail.

The Riemann problem consists of the equation $u_t + Au_x = 0$ together with piecewise constant initial data of the form

$$u(x, 0) = \begin{cases} u_l & x < 0 \\ u_r & x > 0 \end{cases} \quad (6.33)$$

We will assume that this system is **strictly hyperbolic**, which means that the eigenvalues of A are real and distinct, and order them

$$\lambda_1 < \lambda_2 < \cdots < \lambda_m. \quad (6.34)$$

Recall that the general solution to the linear problem is given by (6.12). For the Riemann problem we can simplify the notation if we decompose u_l and u_r as

$$u_l = \sum_{p=1}^m \alpha_p r_p \quad u_r = \sum_{p=1}^m \beta_p r_p. \quad (6.35)$$

Then

$$v_p(x, 0) = \begin{cases} \alpha_p & x < 0 \\ \beta_p & x > 0 \end{cases} \quad (6.36)$$

and so

$$v_p(x, t) = \begin{cases} \alpha_p & \text{if } x - \lambda_p t < 0 \\ \beta_p & \text{if } x - \lambda_p t > 0. \end{cases} \quad (6.37)$$

If we let $P(x, t)$ be the maximum value of p for which $x - \lambda_p t > 0$, then

$$u(x, t) = \sum_{p=1}^{P(x, t)} \beta_p r_p + \sum_{p=P(x, t)+1}^m \alpha_p r_p. \quad (6.38)$$

The determination of $u(x, t)$ at a given point is illustrated in Figure 6.1. In the case shown, $v_1 = \beta_1$ while $v_2 = \alpha_2$ and $v_3 = \alpha_3$. The solution at the point illustrated is thus

$$u(x, t) = \beta_1 r_1 + \alpha_2 r_2 + \alpha_3 r_3. \quad (6.39)$$

Note that the solution is the same at any point in the wedge between the $x' = \lambda_1$ and $x' = \lambda_2$ characteristics. As we cross the p th characteristic, the value of $x - \lambda_p t$ passes through 0 and the corresponding v_p jumps from α_p to β_p . The other coefficients v_i ($i \neq p$) remain constant.

The solution is constant in each of the wedges as shown in Figure 6.2. Across the p th characteristic the solution jumps with the jump given by

$$[u] = (\beta_p - \alpha_p) r_p. \quad (6.40)$$

Note that these jumps satisfy the Rankine-Hugoniot conditions (3.33), since $f(u) = Au$ leads to

$$\begin{aligned} [f] &= A[u] \\ &= (\beta_p - \alpha_p) A r_p \\ &= \lambda_p [u] \end{aligned}$$

and λ_p is precisely the speed of propagation of this jump. The solution $u(x, t)$ in (6.38) can alternatively be written in terms of these jumps as

$$u(x, t) = u_l + \sum_{\lambda_p < x/t} (\beta_p - \alpha_p) r_p \quad (6.41)$$

$$= u_r - \sum_{\lambda_p \geq x/t} (\beta_p - \alpha_p) r_p \quad (6.42)$$

It might happen that the initial jump $u_r - u_l$ is already an eigenvector of A , if $u_r - u_l = (\beta_i - \alpha_i) r_i$ for some i . In this case $\alpha_p = \beta_p$ for $p \neq i$. Then this discontinuity simply propagates with speed λ_i , and the other characteristics carry jumps of zero strength.

In general this is not the case, however, and the jump $u_r - u_l$ cannot propagate as a single discontinuity with any speed without violating the Rankine-Hugoniot condition. We can view “solving the Riemann problem” as finding a way to split up the jump $u_r - u_l$ into a sum of jumps

$$u_r - u_l = (\beta_1 - \alpha_1) r_1 + \cdots + (\beta_m - \alpha_m) r_m, \quad (6.43)$$

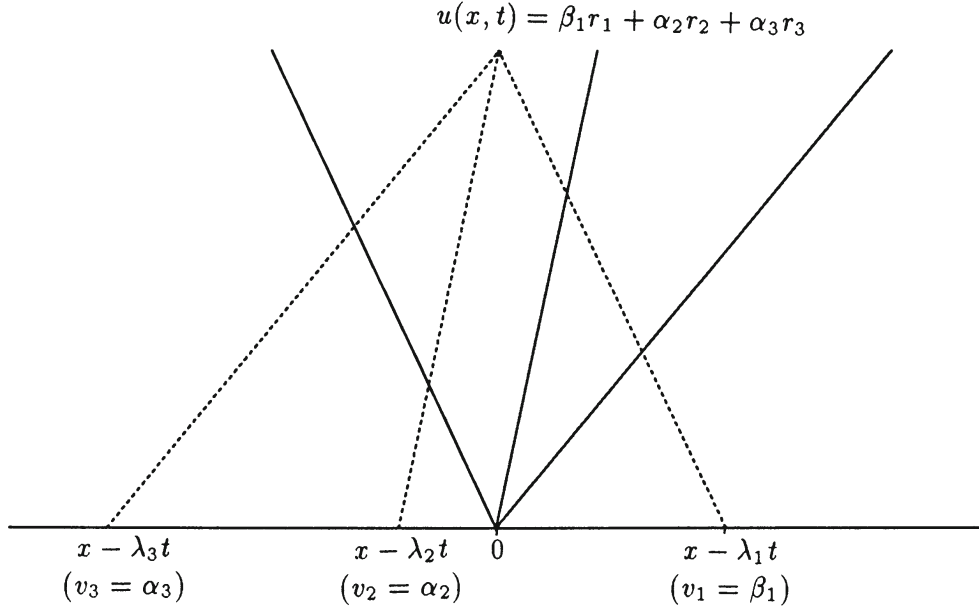


Figure 6.1. Construction of solution to Riemann problem at (x, t) .

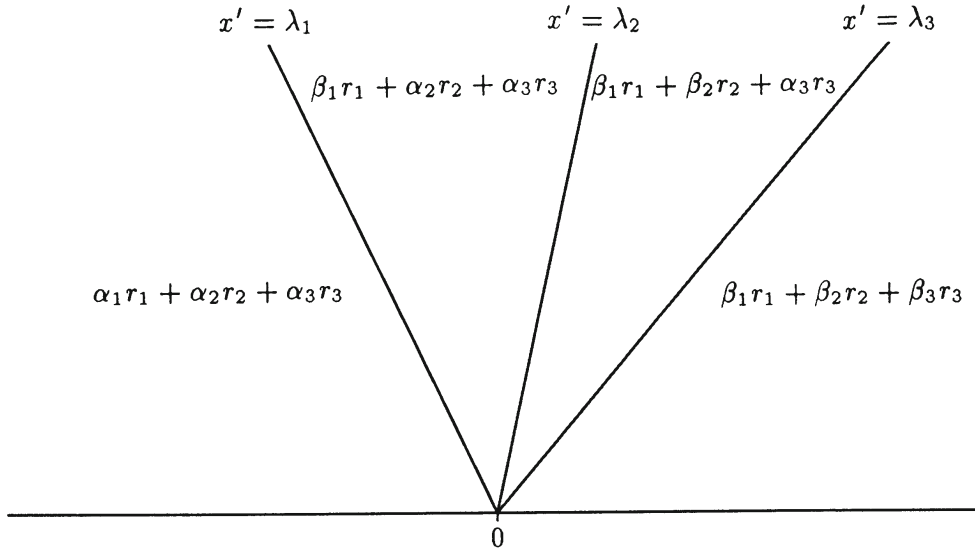


Figure 6.2. Values of solution u in each wedge of $x-t$ plane.

each of which *can* propagate at an appropriate speed λ_i with the Rankine-Hugoniot condition satisfied.

For nonlinear systems we solve the Riemann problem in much the same way: The jump $u_r - u_l$ will usually not have the property that $[f]$ is a scalar multiple of $[u]$, but we can attempt to find a way to split this jump up as a sum of jumps, across each of which this property does hold. (Although life is complicated by the fact that we may need to introduce rarefaction waves as well as shocks.) In studying the solution of the Riemann problem, the jump in the p th family, traveling at constant speed λ_p , is often called the p -wave.

6.5.1 The phase plane

For systems of two equations, it is illuminating to view this splitting in the phase plane. This is simply the u_1 - u_2 plane, where $u = (u_1, u_2)$. Each vector $u(x, t)$ is represented by a point in this plane. In particular, u_l and u_r are points in this plane and a discontinuity with left and right states u_l and u_r can propagate as a single discontinuity only if $u_r - u_l$ is an eigenvector of A , which means that the line segment from u_l to u_r must be parallel to the eigenvector r_1 or r_2 . Figure 6.3 shows an example. For the state u_l illustrated there, the jump from u_l to u_r can propagate as a single discontinuity if and only if u_r lies on one of the two lines drawn through u_l in the direction r_1 and r_2 . These lines give the locus of all points that can be connected to u_l by a 1-wave or a 2-wave. This set of states is called the **Hugoniot locus**. We will see that there is a direct generalization of this to nonlinear systems in the next chapter.

Similarly, there is a Hugoniot locus through any point u_r that gives the set of all points u_l that can be connected to u_r by an elementary p -wave. These curves are again in the directions r_1 and r_2 .

For a general Riemann problem with arbitrary u_l and u_r , the solution consists of two discontinuities travelling with speeds λ_1 and λ_2 , with a new constant state in between that we will call u_m . By the discussion above,

$$u_m = \beta_1 r_1 + \alpha_2 r_2 \quad (6.44)$$

so that $u_m - u_l = (\beta_1 - \alpha_1)r_1$ and $u_r - u_m = (\beta_2 - \alpha_2)r_2$. The location of u_m in the phase plane must be where the 1-wave locus through u_l intersects the 2-wave locus through u_r . This is illustrated in Figure 6.4a.

Note that if we interchange u_r and u_l in this picture, the location of u_m changes as illustrated in Figure 6.4b. In each case we travel from u_l to u_r by first going in the direction r_1 and then in the direction r_2 . This is required by the fact that $\lambda_1 < \lambda_2$ since clearly the jump between u_l and u_m must travel slower than the jump between u_m and u_r if we are to obtain a single-valued solution.

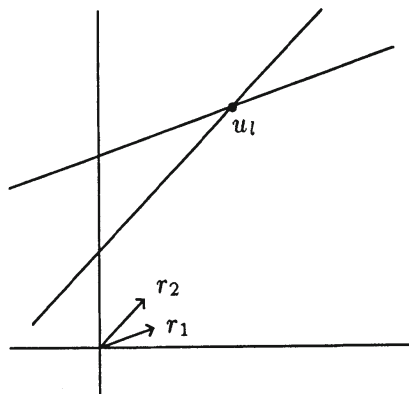


Figure 6.3. The Hugoniot locus of the state u_l consists of all states that differ from u_l by a scalar multiple of r_1 or r_2 .

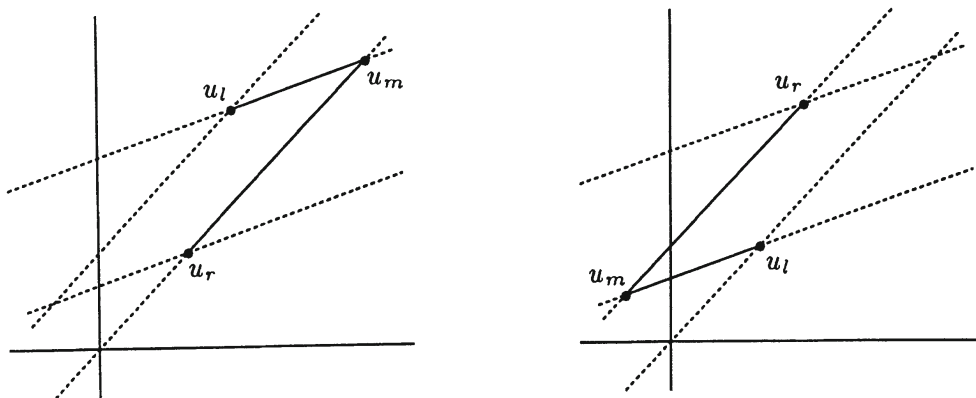


Figure 6.4. The new state u_m arising in the solution to the Riemann problem for two different choices of u_l and u_r .

For systems with more than two equations, the same interpretation is possible but becomes harder to draw since the phase space is now m dimensional. Since the m eigenvectors r_p are linearly independent, we can decompose any jump $u_r - u_l$ into the sum of jumps in these directions, obtaining a piecewise linear path from u_l to u_r in m -dimensional space.

7 Shocks and the Hugoniot Locus

We now return to the nonlinear system $u_t + f(u)_x = 0$, where $u(x, t) \in \mathbb{R}^m$. As before we assume strict hyperbolicity, so that $f'(u)$ has distinct real eigenvalues $\lambda_1(u) < \cdots < \lambda_m(u)$ and hence linearly independent eigenvectors. We choose a particular basis for these eigenvectors, $\{r_p(u)\}_{p=1}^m$, usually chosen to be normalized in some manner, e.g. $\|r_p(u)\| \equiv 1$.

In the previous chapter we constructed the solution to the general Riemann problem for a linear hyperbolic system of conservation laws. Our goal in the next two chapters is to perform a similar construction for the nonlinear Riemann problem. In the linear case the solution consists of m waves, which are simply discontinuities traveling at the characteristic velocities of the linear system. In the nonlinear case our experience with the scalar equation leads us to expect more possibilities. In particular, the physically relevant vanishing viscosity solution may contain rarefaction waves as well as discontinuities. In this chapter we will first ignore the entropy condition and ask a simpler question: is it possible to construct a weak solution of the Riemann problem consisting only of m discontinuities propagating with constant speeds $s_1 < s_2 < \cdots < s_m$? As we will see, the answer is yes for $\|u_l - u_r\|$ sufficiently small.

7.1 The Hugoniot locus

Recall that if a discontinuity propagating with speed s has constant values $\hat{u}$ and $\tilde{u}$ on either side of the discontinuity, then the Rankine-Hugoniot jump condition must hold,

$$f(\tilde{u}) - f(\hat{u}) = s(\tilde{u} - \hat{u}). \quad (7.1)$$

Now suppose we fix the point $\hat{u} \in \mathbb{R}^m$ and attempt to determine the set of all points $\tilde{u}$ which can be connected to $\hat{u}$ by a discontinuity satisfying (7.1) for some s . This gives a system of m equations in $m + 1$ unknowns: the m components of $\tilde{u}$, and s . This leads us to expect one parameter families of solutions.

We know that in the linear case there are indeed m such families for any $\hat{u}$. In the p th family the jump $\tilde{u} - \hat{u}$ is some scalar multiple of r_p , the p th eigenvector of A . We can

parameterize these families of solutions using this scalar multiple, say ξ , and we obtain the following solution curves:

$$\begin{aligned}\tilde{u}_p(\xi; \hat{u}) &= \hat{u} + \xi r_p \\ s_p(\xi; \hat{u}) &= \lambda_p\end{aligned}$$

for $p = 1, 2, \dots, m$. Note that $\tilde{u}_p(0; \hat{u}) = \hat{u}$ for each p and so through the point $\hat{u}$ in phase space there are m curves (straight lines in fact) of possible solutions. This is illustrated in Figure 6.4 for the case $m = 2$. The two lines through each point are the states that can be connected by a discontinuity with jump proportional to r_1 or r_2 .

In the nonlinear case we also obtain m curves through any point $\hat{u}$, one for each characteristic family. We again parameterize these curves by $\tilde{u}_p(\xi; \hat{u})$ with $\tilde{u}_p(0; \hat{u}) = \hat{u}$ and let $s_p(\xi; \hat{u})$ denote the corresponding speed. To simplify notation, we will frequently write these as simply $\tilde{u}_p(\xi)$, $s_p(\xi)$ when the point $\hat{u}$ is clearly understood.

The Rankine-Hugoniot condition gives

$$f(\tilde{u}_p(\xi)) - f(\hat{u}) = s_p(\xi)(\tilde{u}_p(\xi) - \hat{u}). \quad (7.2)$$

Differentiating this expression with respect to ξ and setting $\xi = 0$ gives

$$f'(\hat{u})\tilde{u}'_p(0) = s_p(0)\tilde{u}'_p(0) \quad (7.3)$$

so that $\tilde{u}'_p(0)$ must be a scalar multiple of the eigenvector $r_p(\hat{u})$ of $f'(\hat{u})$, while $s_p(0) = \lambda_p(\hat{u})$. The curve $\tilde{u}_p(\xi)$ is thus tangent to $r_p(\hat{u})$ at the point $\hat{u}$. We have already observed this, by a slightly different argument, in discussing weak shocks in Chapter 6. For a system of $m = 2$ equations, this is easily illustrated in the phase plane. An example for the isothermal equations of gas dynamics is discussed below, see Figure 7.1.

For smooth f , it can be shown using the implicit function theorem that these solution curves exist locally in a neighborhood of $\hat{u}$, and that the functions $\tilde{u}_p$ and s_p are smooth. See Lax[45] or Smoller[77] for details. These curves are called Hugoniot curves. The set of all points on these curves is often collectively called the **Hugoniot locus** for the point $\hat{u}$. If $\tilde{u}_p$ lies on the p th Hugoniot curve through $\hat{u}$, then we say that $\hat{u}$ and $\tilde{u}_p$ are connected by a p -shock.

EXAMPLE 7.1. The isothermal equations of gas dynamics (5.32) provide a relatively simple example of the nonlinear theory.

If we let m represent the momentum, $m = \rho v$, then the system becomes

$$\begin{aligned}\rho_t + m_x &= 0 \\ m_t + (m^2/\rho + a^2\rho)_x &= 0\end{aligned} \quad (7.4)$$

or $u_t + f(u)_x = 0$ where $u = (\rho, m)$.

The Jacobian matrix is

$$f'(u) = \begin{bmatrix} 0 & 1 \\ a^2 - m^2/\rho^2 & 2m/\rho \end{bmatrix}. \quad (7.5)$$

The eigenvalues are

$$\lambda_1(u) = m/\rho - a, \quad \lambda_2(u) = m/\rho + a \quad (7.6)$$

with eigenvectors

$$r_1(u) = \begin{bmatrix} 1 \\ m/\rho - a \end{bmatrix}, \quad r_2(u) = \begin{bmatrix} 1 \\ m/\rho + a \end{bmatrix}. \quad (7.7)$$

These could be normalized but it is easiest to leave them in this simple form.

Now let's fix a state $\hat{u}$ and determine the set of states $\tilde{u}$ that can be connected by a discontinuity. The Rankine-Hugoniot condition (7.1) becomes, for this system,

$$\begin{aligned} \tilde{m} - \hat{m} &= s(\tilde{\rho} - \hat{\rho}) \\ (\tilde{m}^2/\tilde{\rho} + a^2\tilde{\rho}) - (\hat{m}^2/\hat{\rho} + a^2\hat{\rho}) &= s(\tilde{m} - \hat{m}). \end{aligned} \quad (7.8)$$

This gives two equations in the three unknowns $\tilde{\rho}$, $\tilde{m}$, and s . These equations can be easily solved for $\tilde{m}$ and s in terms of $\tilde{\rho}$, giving

$$\tilde{m} = \tilde{\rho}\hat{m}/\hat{\rho} \pm a\sqrt{\tilde{\rho}/\hat{\rho}} (\tilde{\rho} - \hat{\rho}) \quad (7.9)$$

and

$$s = \hat{m}/\hat{\rho} \pm a\sqrt{\tilde{\rho}/\hat{\rho}}. \quad (7.10)$$

The $\pm$ signs in these equations give two solutions, one for each family. Since $\tilde{m}$ and s can be expressed in terms of $\tilde{\rho}$, we can easily parameterize these curves by taking, for example,

$$\tilde{\rho}_p(\xi; \hat{u}) = \hat{\rho}(1 + \xi), \quad p = 1, 2. \quad (7.11)$$

We then have

$$\tilde{u}_1(\xi; \hat{u}) = \hat{u} + \xi \begin{bmatrix} \hat{\rho} \\ \hat{m} - a\hat{\rho}\sqrt{1+\xi} \end{bmatrix}, \quad s_1(\xi; \hat{u}) = \hat{m}/\hat{\rho} - a\sqrt{1+\xi}. \quad (7.12)$$

and

$$\tilde{u}_2(\xi; \hat{u}) = \hat{u} + \xi \begin{bmatrix} \hat{\rho} \\ \hat{m} + a\hat{\rho}\sqrt{1+\xi} \end{bmatrix}, \quad s_2(\xi; \hat{u}) = \hat{m}/\hat{\rho} + a\sqrt{1+\xi}. \quad (7.13)$$

The choice of signs for each family is determined by the behavior as $\xi \rightarrow 0$, where the relation (7.3) must hold. It is easy to check that with the above choice we have

$$\begin{aligned} \frac{\partial}{\partial \xi} \tilde{u}_p(0; \hat{u}) &= \hat{\rho} r_p(\hat{u}) \propto r_p(\hat{u}), \\ s_p(0; \hat{u}) &= \lambda_p(\hat{u}), \end{aligned}$$

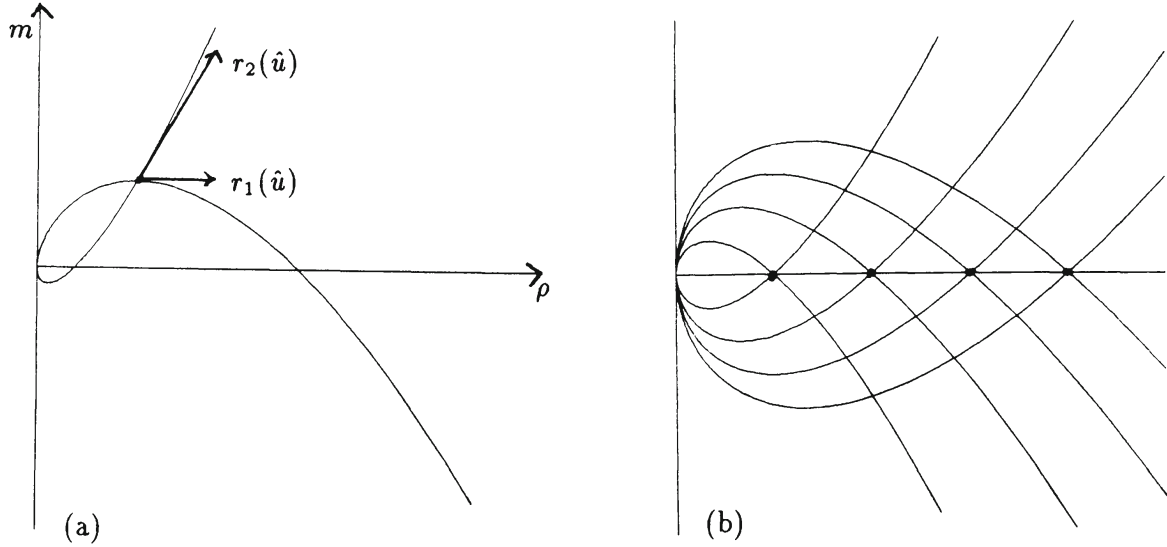


Figure 7.1. a) Hugoniot locus for the state $\hat{u} = (1, 1)$ in the isothermal gas dynamics equations with $a = 1$. b) Variation of these curves for $\hat{u} = (\hat{\rho}, 0)$, $\hat{\rho} = 1, 2, 3, 4$.

as expected.

Notice that real-valued solutions exist only for $\xi > -1$ and that $\tilde{u}_p(-1; \hat{u}) = (0, 0)$ for $p = 1, 2$ and any $\hat{u}$. Thus, each Hugoniot locus terminates at the origin (the *vacuum state*, since $\rho = 0$). There are no states with $\rho < 0$ that can be connected to $\hat{u}$ by a propagating discontinuity. The curves $\tilde{u}_p(\xi)$ are illustrated in Figure 7.1a for one particular choice of $\hat{u}$ and $a = 1$. Figure 7.1b shows how these curves vary with $\hat{u}$. In this case we see $\tilde{u}_p(\xi; \hat{u})$ for $\hat{u} = (\hat{\rho}, 0)$, $\hat{\rho} = 1, 2, 3, 4$.

EXERCISE 7.1. Determine the Hugoniot locus for the shallow water equations (5.38).

7.2 Solution of the Riemann problem

Now suppose that we wish to solve the Riemann problem with left and right states u_l and u_r (and recall that we are ignoring the entropy condition at this point). Just as in the linear case, we can accomplish this by finding an intermediate state u_m such that u_l and u_m are connected by a discontinuity satisfying the Rankine-Hugoniot condition, and so are u_m and u_r . Graphically we accomplish this by drawing the Hugoniot locus for each of the states u_l and u_r and looking for intersections. See Figure 7.2 for an example with the isothermal equations.

In this example there are two points of intersection, labelled u_m and u_m^* , but only u_m gives a single-valued solution to the Riemann problem since we need the jump from

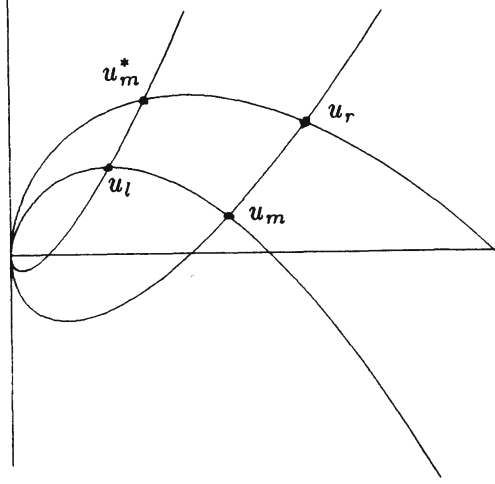


Figure 7.2. Construction of a weak solution to the Riemann problem with left and right states u_l and u_r .

u_l to u_m to travel more slowly than the jump from u_m to u_r . This requires that u_m be connected to u_l by a 1-shock while u_r is connected to u_m by a 2-shock, due to our convention that $\lambda_1(u) < \lambda_2(u)$. Of course our requirement really involves the shock speeds, not the characteristic speeds, but note that any 1-shock connected to u_m has speed

$$s_1(\xi; u_m) = m_m/\rho_m - a\sqrt{1 + \xi/\rho_m} < m_m/\rho_m \quad \forall \xi > -\rho_m$$

while any 2-shock has speed

$$s_2(\xi; u_m) = m_m/\rho_m + a\sqrt{1 + \xi/\rho_m} > m_m/\rho_m \quad \forall \xi > -\rho_m$$

and consequently $s_1(\xi_1; u_m) < s_2(\xi_2; u_m)$ for all $\xi_1, \xi_2 > -\rho_m$.

The state u_m can be found algebraically by using our explicit expressions for the Hugoniot locus. We wish to find a state (ρ_m, m_m) which is connected to u_l by a 1-shock and to u_r by a 2-shock. Consequently equation (7.9) with the minus sign should hold for $\tilde{u} = u_m, \hat{u} = u_l$, and the same equation with the plus sign should hold for $\tilde{u} = u_m, \hat{u} = u_r$. Equating the two resulting expressions for m_m gives

$$\rho_m m_l / \rho_l - a\sqrt{\rho_m / \rho_l} (\rho_m - \rho_l) = \rho_m m_r / \rho_r + a\sqrt{\rho_m / \rho_r} (\rho_m - \rho_r). \quad (7.14)$$

Setting $z = \sqrt{\rho_m}$, we obtain a quadratic equation for z ,

$$\left(\frac{a}{\sqrt{\rho_r}} + \frac{a}{\sqrt{\rho_l}} \right) z^2 + \left(\frac{m_r}{\rho_r} - \frac{m_l}{\rho_l} \right) z - a(\sqrt{\rho_r} + \sqrt{\rho_l}) = 0. \quad (7.15)$$

This equation has a unique positive solution $\bar{z}$, and $\rho_m = \bar{z}^2$. We can then determine m_m by evaluating either side of (7.14).

More generally, for a system of m equations we can attempt to solve the Riemann problem by finding a sequence of states $u_1, u_2, \dots, u_{m-1}$ such that u_l is connected to u_1 by a 1-shock, u_1 is connected to u_2 by a 2-shock, and so on, with u_{m-1} connected to u_r by an m -shock. If u_l and u_r are sufficiently close together then this can always be achieved. Lax proved a stronger version of this in his fundamental paper [44]. (By considering rarefaction waves also, the entropy satisfying solution can be constructed in a similar manner.) See also Lax[45] and Smoller[77] for detailed proofs. The idea is quite simple and will be summarized here following Lax[45]. From u_l we know that we can reach a one parameter family of states $u_1(\xi_1)$ through a 1-shock. From $u_1(\xi_1)$ we can reach another one parameter family of states $u_2(\xi_1, \xi_2)$ through a 2-shock. Continuing, we see that from u_l we can reach an m parameter family of states $u_m(\xi_1, \xi_2, \dots, \xi_m)$. Moreover, we know that

$$\left. \frac{\partial u_m}{\partial \xi_p} \right|_{\xi_1=\dots=\xi_m=0} \propto r_p(u_l), \quad p = 1, 2, \dots, m.$$

These vectors are linearly independent by our hyperbolicity assumption, and hence the Jacobian of the mapping $u_m : \mathbb{R}^m \rightarrow \mathbb{R}^m$ is nonsingular. It follows that the mapping u_m is bijective in a neighborhood of the origin. Hence for any u_r sufficiently close to u_l there is a unique set of parameters $\xi_1, \dots, \xi_m$ such that $u_r = u_m(\xi_1, \dots, \xi_m)$.

7.2.1 Riemann problems with no solution

For a general nonlinear system the local result quoted above is the best we can expect, and there are examples where the Riemann problem has no weak solution when $\|u_l - u_r\|$ is sufficiently large. In such examples the shock curves simply do not intersect in the required way. One such example is given by Smoller[76].

For specific systems of conservation laws one can often prove that solutions to the Riemann problem exist more globally. For example, in the isothermal case we can clearly always solve the Riemann problem provided $\rho_l, \rho_r > 0$. The case where ρ_l or ρ_r is negative is of little interest physically, but even then the Riemann problem can be solved by taking the vacuum state $u_m = (0, 0)$ as the intermediate state. We encounter problems only if $\rho = 0$ and $m \neq 0$ for one of the states, but since the equations involve m/ρ this is not surprising.

7.3 Genuine nonlinearity

In defining the Hugoniot locus above, we ignored the question of whether a given discontinuity is physically relevant. The state $\tilde{u}$ is in the Hugoniot locus of $\hat{u}$ if the jump

satisfies the Rankine-Hugoniot condition, regardless of whether this jump could exist in a vanishing viscosity solution. We would now like to define an entropy condition that can be applied directly to a discontinuous weak solution to determine whether the jumps should be allowed. In Chapter 3 we considered several forms of the entropy condition for scalar equations. In the convex case, the simplest condition is simply that characteristics should go *into* the shock as time advances, resulting in the requirement (3.45),

$$f'(u_l) > s > f'(u_r).$$

Lax[44] proposed a simple generalization of this entropy condition to systems of equations that are genuinely nonlinear, a natural generalization of the convex scalar equation. The p th characteristic field is said to be **genuinely nonlinear** if

$$\nabla \lambda_p(u) \cdot r_p(u) \neq 0 \quad \text{for all } u, \quad (7.16)$$

where $\nabla \lambda_p(u) = (\partial \lambda_p / \partial u_1, \dots, \partial \lambda_p / \partial u_m)$ is the gradient of $\lambda_p(u)$. Note that in the scalar case, $m = 1$ and $\lambda_1(u) = f'(u)$ while $r_1(u) = 1$ for all u . The condition (7.16) reduces to the convexity requirement $f''(u) \neq 0 \ \forall u$. This implies that the characteristic speed $f'(u)$ is monotonically increasing or decreasing as u varies, and leads to a relatively simple solution of the Riemann problem.

For a system of equations, (7.16) implies that $\lambda_p(u)$ is monotonically increasing or decreasing as u varies along an integral curve of the vector field $r_p(u)$. These integral curves will be discussed in detail in the next chapter, where we will see that through a rarefaction wave u varies along an integral curve. Since monotonicity of the propagation speed λ_p is clearly required through a rarefaction wave, genuine nonlinearity is a natural assumption.

7.4 The Lax entropy condition

For a genuinely nonlinear field, Lax's entropy condition says that a jump in the p th field (from u_l to u_r , say) is admissible only if

$$\lambda_p(u_l) > s > \lambda_p(u_r) \quad (7.17)$$

where s is again the shock speed. Characteristics in the p th family disappear into the shock as time advances, just as in the scalar case.

EXAMPLE 7.2. For isothermal gas dynamics we can easily verify that both fields are genuinely nonlinear. Since $\lambda_p = m/\rho \pm a$, we compute that in each case

$$\nabla \lambda_p(u) = \begin{bmatrix} -m/\rho^2 \\ 1/\rho \end{bmatrix}, \quad p = 1, 2. \quad (7.18)$$

Using (7.7), we compute that

$$\nabla \lambda_1(u) \cdot r_1(u) = -a/\rho \quad (7.19)$$

$$\nabla \lambda_2(u) \cdot r_2(u) = a/\rho. \quad (7.20)$$

These quantities are both nonzero for all u .

Now suppose u_l and u_r are connected by a 1-shock. Then u_l lies in the Hugoniot locus of u_r and also u_r lies in the Hugoniot locus of u_l . We can thus evaluate the shock speed s using (7.10) (with the minus sign, since the jump is a 1-shock) in two different ways, obtaining

$$s = \frac{m_l}{\rho_l} - a\sqrt{\rho_r/\rho_l} = \frac{m_r}{\rho_r} - a\sqrt{\rho_l/\rho_r}. \quad (7.21)$$

Since $\lambda_1(u) = m/\rho - a$, the entropy condition (7.17) becomes

$$\frac{m_l}{\rho_l} - a > \frac{m_l}{\rho_l} - a\sqrt{\rho_r/\rho_l} = \frac{m_r}{\rho_r} - a\sqrt{\rho_l/\rho_r} > \frac{m_r}{\rho_r} - a \quad (7.22)$$

and is clearly satisfied if and only if $\rho_r > \rho_l$.

Notice that since the fluid velocity is $v = m/\rho$, 1-shocks always travel slower than the fluid on either side, and so a given fluid particle passes through the shock from left to right (i.e. its state jumps from u_l to u_r). A consequence of the entropy condition is that the density of the gas must *increase* as it goes through the shock. This is also true more generally in the full Euler equations. The gas can only be compressed as the shock passes, not rarefied (rarefaction occurs, naturally enough, through a rarefaction wave rather than a shock).

For 2-shocks the entropy condition requires

$$\frac{m_l}{\rho_l} + a > \frac{m_l}{\rho_l} + a\sqrt{\rho_r/\rho_l} = \frac{m_r}{\rho_r} + a\sqrt{\rho_r/\rho_l} > \frac{m_r}{\rho_r} + a \quad (7.23)$$

which is now satisfied only if $\rho_r < \rho_l$. But note that 2-shocks travel faster than the fluid on either side, so that particles pass through the shock from right to left. So the entropy condition has the same physical interpretation as before: the density must jump from ρ_r to a higher value ρ_l as the gas goes through the shock.

We can now reconsider the Hugoniot locus of a point $\hat{u}$ and retain only the points $\tilde{u}$ that can be connected to $\hat{u}$ by an entropy-satisfying shock, discarding the entropy-violating shocks. In order to do this, we must first decide whether $\hat{u}$ is to lie to the left of the discontinuity or to the right. The entropy condition (7.17), unlike the Rankine-Hugoniot condition, is not symmetric in the two states.

Figure 7.3a shows the set of states that can be connected to the right of a given state $\hat{u}$ by an entropy-satisfying shock. Figure 7.3b shows the set of states that can be connected to the left of the same state $\hat{u}$. Note that the union of these curves gives the full Hugoniot locus. Each branch of the Hugoniot locus splits into two parts at $\hat{u}$; states on one side can only be connected to the left, states on the other side can only be connected to the right.

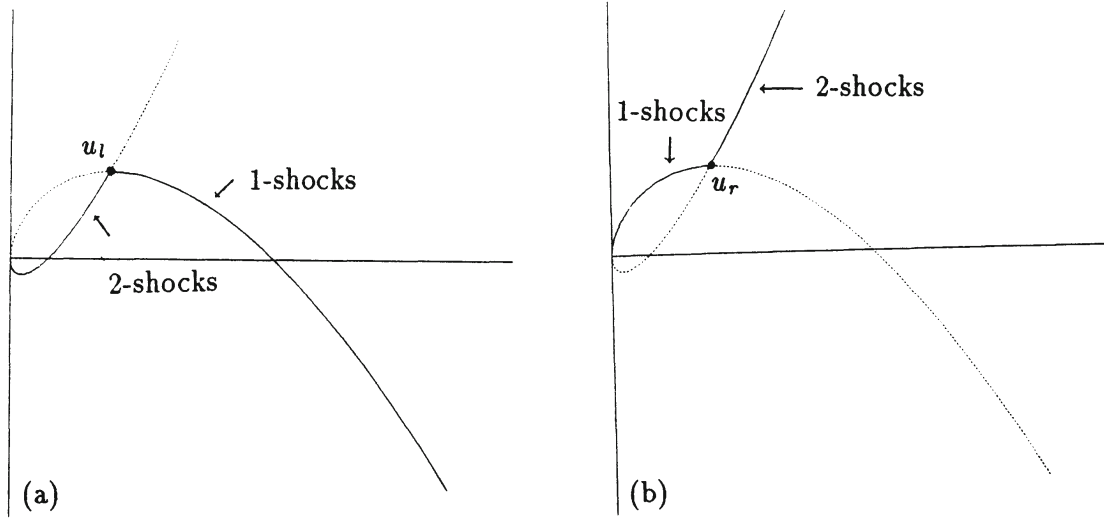


Figure 7.3. a) States u_r that can be connected to $u_l = \hat{u}$ by an entropy-satisfying shock. b) States u_l that can be connected to $u_r = \hat{u}$ by an entropy-satisfying shock. In each case the entropy-violating branches of the Hugoniot locus are shown as dashed lines.

7.5 Linear degeneracy

The assumption of genuine nonlinearity is obviously violated for a constant coefficient linear system, in which $\lambda_p(u)$ is constant and hence $\nabla \lambda_p \equiv 0$. More generally, for a nonlinear system it might happen that in one of the characteristic fields the eigenvalue $\lambda_p(u)$ is constant along integral curves of this field, and hence

$$\nabla \lambda_p(u) \cdot r_p(u) \equiv 0 \quad \forall u. \quad (7.24)$$

(Of course the value of $\lambda_p(u)$ might vary from one integral curve to the next.) In this case we say that the p th field is **linearly degenerate**. This may seem rather unlikely, and not worth endowing with a special name, but in fact the Euler equations have this property. We will see later that for this system of three equations, two of the fields are genuinely nonlinear while the third is linearly degenerate.

A discontinuity in a linearly degenerate field is called a **contact discontinuity**. This name again comes from gas dynamics. In a shock tube problem the gas initially on one side of the diaphragm never mixes with gas from the other side (in the inviscid Euler equations). As time evolves these two gases remain in contact along a ray in the x - t plane along which, in general, there is a jump in density. This is the contact discontinuity.

For general systems, if the p th field is linearly degenerate and u_l and u_r are connected by a discontinuity in this field, then it can be shown that u_l and u_r lie on the same integral curve of $r_p(u)$, so that $\lambda_p(u_l) = \lambda_p(u_r)$. Moreover, the propagation speed s is also equal to

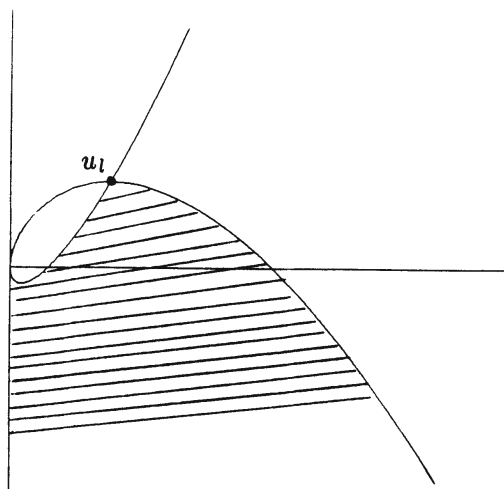


Figure 7.4. The shaded region shows the set of points u_r for which the Riemann problem with states u_l , u_r will have an entropy satisfying solution consisting of two shocks.

$\lambda_p(u_l)$. Consequently, the p -characteristics are parallel to the propagating discontinuity on each side, just as in the case of a linear system. Contact discontinuities can occur in vanishing viscosity solutions, and so for systems with both genuinely nonlinear and linearly degenerate fields, we should modify the entropy condition (7.17) to read

$$\lambda_p(u_l) \geq s \geq \lambda_p(u_r). \quad (7.25)$$

7.6 The Riemann problem

Returning to the Riemann problem, we see that for general data u_l and u_r , the weak solution previously constructed will not be the physically correct solution if any of the resulting shocks violate the entropy condition. In this case we need to also use rarefaction waves in our construction of the solution. This will be done in the next chapter.

EXAMPLE 7.3. In particular, the weak solution to the isothermal equations constructed in Figure 7.2 is nonphysical, since the 2-shock does not satisfy the entropy condition. For the state u_l illustrated there, a physically correct Riemann solution consisting of two shocks will exist only if the state u_r lies in the region shaded in Figure 7.4.

EXERCISE 7.2. Consider the shallow water equations (5.38).

1. Show that a weak solution to the Riemann problem consisting only of shocks always exists if $\varphi_l, \varphi_r > 0$. Determine the intermediate state u_m for given states u_l and u_r .